



Methyl butanoate inhibition of n-heptane diffusion flames through an evaluation of transport and chemical kinetics

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ABSTRACT

The extinction limits of methyl butanoate, n-heptane, and methyl butanoate/n-heptane diffusion flames have been measured as a function of fuel mole fraction with nitrogen dilution in counterflow with air. On a mole fraction basis, methyl butanoate diffusion flames are observed to have a much lower extinction strain rate than n-heptane diffusion flames and the extinction strain rate of n-heptane/methyl butanoate diffusion flames is observed to increase significantly as the n-heptane fraction is increased.

Based on previous works, detailed chemical kinetic models to describe the high temperature oxidation of these fuel mixtures are assembled, tested and reduced. When the transport properties of ester species are re-evaluated by means of a thorough literature review, numerical computations of extinction generally reproduce experimental results for the pure fuels as well as for mixtures. An in-depth analysis of the kinetic model computations reveals that the extinction behaviour of both fuels is due to (1) fuel energy content affects and (2) the chemical kinetic potential of each fuel to produce the hydroperoxy radical. Comparatively, in n-heptane flames reactive ethyl radicals and ethylene are the major intermediates formed, but in methyl butanoate flames the major intermediates are formyl radicals and formaldehyde. In all flames studied, increased strain rates affect an increased interaction of formyl and/or vinyl radicals with molecular oxygen leading to a transition from hydrogen atom production at low strain rates, to the production of large quantities of the hydroperoxy radical at higher strain rates. The formation of the hydroperoxy radical induces extinction in each flame by directly interfering with the important radical chain branching and exothermic elementary reactions of H atoms and OH radicals that are dominant in weakly strained flames.

It is postulated that the similar inhibitive effect of methyl butanoate fuelled flames will also be observed for more biodiesel like, larger n-alkyl esters when compared to equivalent n-alkanes. The diffusive extinction limits of methyl decanoate diffusion flames are also measured and show reactivity comparable to n-heptane diffusion flames by a molar comparison.

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1. Introduction

Biofuels have become an important source of transportation fuel helping to minimize climate change and provide energy security for otherwise energy dependent nations. For use within existing engines and distribution systems, biofuels are being blended with petroleum based transportation fuels [1,2]. Therefore, the understanding and quantitative prediction of burning and emission properties of biofuel mixtures with gasoline or diesel fuels are important to future fuel and engine designs. In addition to ethanol, one of the most prevalent biofuels is methyl ester biodiesel.

Commercially, biodiesel is produced by the transesterification of the oils harvested from, for example, rapeseed and soybean. Recently, the possibility of producing larger quantities of biodiesel

by the growing of algae is emerging. Like the vegetable oils, algae oil is composed of triglycerides that must be transesterified with the use of methanol to produce methyl ester biodiesel. The much higher yield per unit land, as well as the prospect of using non agricultural lands for algae production has led to a renewed interest in the practical consideration of biodiesel as a significant future transportation fuel stock [3].

Over the last 10 years, substantial research within the combustion community has significantly elucidated the chemical kinetic processes of methyl ester fuel oxidation, particularly through the study of model molecules such as methyl butanoate and more recently methyl decanoate. Fisher et al. [4] initially constructed a kinetic model for the much smaller than real biodiesel methyl ester, methyl butanoate (MB). Subsequent studies for example, [5–10], have empirically characterised this fuel and refined the chemical kinetic description of its oxidation. These works have shown that although MB is deficient as a biodiesel surrogate at low temperatures

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(<900 K) [6,7], observations at high temperatures provide an accurate representation of the chemical kinetic processes associated with methyl ester combustion. The methyl ester chemical functionality differentiates the molecular structure of biodiesels from conventional petroleum derived long chain hydrocarbons. Due to the deficiencies of MB as a low temperature biodiesel surrogate, several larger biodiesel surrogates have recently been proposed, and a kinetic model has been constructed but remains largely untested for methyl decanoate [11].

Although there have been some significant advancements in the description of methyl ester chemical kinetics, there still exists a lack of fundamental characterisation of the role of the methyl ester functionality in oxidation, particularly under practically important flame propagation and stabilisation conditions. While a clearer understanding of methyl ester oxidation chemistry continues to emerge, there has been very little detailed characterisation or fundamental analysis of the interaction of methyl ester components with those types of organic structures found in petroleum-derived fuels or Fischer–Tropsch produced alternative fuels.

Fundamental strained extinction limits of diffusion flames are relevant to the performance of practical combustion systems as they define the existence of very high temperature regions that can lead to the formation of NO_x in non-premixed flames. Therefore, it is important to evaluate the extinction limits of fuels and the impact of fuel blending on extinction limits. In the present study the effects of the methyl ester moiety on the extinction limits of fossil fuel–biofuel mixtures is fundamentally investigated. The counterflow burner method is used to determine diffusion flame extinction limits as a function of fuel mole fraction for n-heptane (nH), the most well understood model biodiesel, methyl butanoate (MB) and for two n-heptane/methyl butanoate blends. n-heptane is chosen as it is a chemical kinetically well understood alkane representative of the normal or lightly branched hydrocarbons found in petroleum-derived and Fischer–Tropsch derived alternative fuels. In addition, to examine how these observations translate to those that would be made for real biodiesels, the extinction limits of the larger model biodiesel, methyl decanoate (MD) are also determined.

In order to fundamentally interpret the experimental observations made in this study and to explain the flame combustion mechanism of n-heptane/methyl butanoate blends, detailed chemical kinetic models to describe the oxidation of both fuels are necessarily assembled. Testing of these models against experiment and subsequent analysis provides fundamental insight into the effect of the methyl ester functional group on diffusion flame reactivity, and provides instruction on what aspects of methyl ester combustion modelling require improvement.

2. Experimental and numerical methods

The experimental apparatus for the measurement of diffusion flame extinction limits consists of a counterflow burner integrated with a liquid fuel vaporisation system and flow controllers as described previously [12,13]. A schematic of the apparatus is shown in Fig. 1. The counterflow burner is 13 mm in internal diameter at the exit of the converged nozzle, resulting in a uniform velocity profile at the nozzle exit. The burner separation distance is maintained at 9 mm. A chamber of 300 cm³ allows for reduced flow response time to enhance the vaporisation of low vapour pressure liquids. A steady state atomisation is achieved by injecting liquid fuel and heated nitrogen into the vaporisation chamber through an inner nozzle of 0.2 mm and outer co-flow nozzle of 1 mm respectively. The temperature at the exit of the vaporisation chamber is maintained at 500 K with PID control. The upper burner is also kept at 500 ± 5 K. The oxidizer temperature is 298 K. Fuel flow rates, nH (>99% purity), MB (>99%), MD (>99%), were volumet-

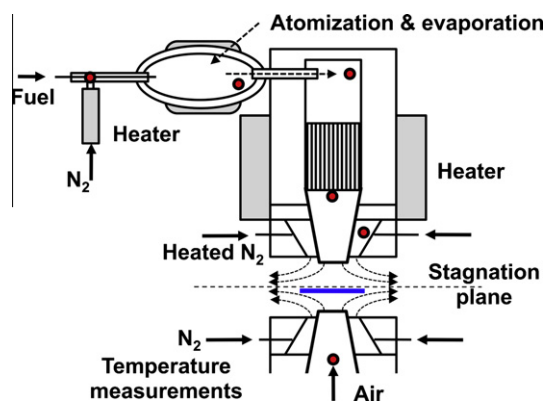


Fig. 1. Schematic of experimental setup, red circles indicates heated zones. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rically administered by a syringe pump (Harvard Apparatus, PHD 22/2000). For MB/nH blended fuel experiments, the fuels are measured on a mass basis and premixed in liquid form in a stainless steel syringe before evaporation. Extinction limits are measured by gradually increasing the strain rates with equal momentum for both streams until flame extinction occurs [12,13]. Global strain rates at the oxidizer side were used by accounting for the density difference between fuel and oxidizer sides by Eq. (1) [14].

$$a = \frac{2U_o}{L} \left(1 + \frac{U_f}{U_o} \sqrt{\frac{\rho_f}{\rho_o}} \right) \quad (1)$$

Here, ρ represents density, U the flow velocity at the nozzle exit, L is the distance between fuel and oxidizer nozzles, and subscript f and o indicate fuel and oxidizer respectively.

The radial velocity profiles at the nozzle exits need to be uniform with a “top-hat” profile [15], so that exit boundary conditions are appropriate for one dimensional numerical modelling. A one dimensional approximation is essential to analysing problems that involve large detailed kinetic models. Velocity profiles at the nozzle exit were measured with particle imaging velocimetry for the range of flow velocities used to achieve extinction in the current study. These tests revealed that the current converged nozzle provides a “top-hat” velocity profile with less than 3% non-uniformity.

In order to accurately characterize the global strain, the flow at the nozzle exit must be unperturbed by the gas expansion or buoyancy effect from the reaction zone. The buoyancy effect may be assessed by evaluating the Richardson number, Ri , defined as $\Delta\rho g L / \rho_f U_f^2$. Where $\Delta\rho$ is the density difference between unburned and burned gases and g is the gravitational force. For the conditions of this experimental study, Ri becomes larger than unity at $a < 100$ 1/s, and decreases to 0.06 at $a = 400$ 1/s. Since large values of Ri indicate potential buoyancy effects, measurements with extinction strain rates below 100 1/s were excluded from the analysis. The definition of global strain rate is derived based on the thin reaction zone approximation with the asymptotic approach of Seshadri and Williams [14]. Thus, the validity of utilising the global strain rate must be confirmed in terms of the separation distance between two nozzles. Extinction limits were measured by varying the separation distance from 9 to 13 mm. Consistent results were observed within 3% uncertainty, confirming that it is appropriate to use global strain rate for comparison of extinction measurements and numerical calculations.

Numerical calculations are performed using the OPPDIF module of the CHEMKIN package [16] employing plug flow boundary conditions, to account for the radial velocity profile. Extinction strain rates computed with the mixture averaged transport description

are found to be $10\text{--}15\text{ s}^{-1}$ higher than those computed by the multicomponent method, and therefore fall within the global uncertainties described below. Further details of these numerical calculations are described elsewhere [12,13]. All experimental measurements are available in the [Supplementary Materials](#).

2.1. Kinetic model constructions

It is the principle objective of this study to fundamentally evaluate any differences which may be particular to the high temperature combustion phenomena of biodiesel type fuels relative to those of fossil derived fuels. This requires a detailed numerical evaluation of the combustion chemistry occurring in flames of such representative fuels. Thus in order to fundamentally interpret the extinction limit observations for nH/MB flames, a kinetic model that describes the oxidation of mixtures of the two fuels is required.

2.1.1. Model I

To this end we have taken the updated n-heptane kinetic model of Curran et al. 2004 [17] that has previously been tested under an array of conditions, amongst these, some very similar to those explored in this study [18]. To it, the MB sub-mechanism that has recently been developed and tested by Dooley et al. [7], stemming from the work of Fisher et al., has been added. The small species chemistry adopted in the work of Dooley et al. is that of Petersen et al. [19], and has known short comings under flame conditions due to apparent discrepancies in elementary rate constants with employed thermochemical parameters. Consequently, here we have chosen to maintain the $\text{C}_0\text{--}\text{C}_4$ chemistry utilised by Curran et al. for n-heptane oxidation in the combined model for nH/MB referred to as *Model I*. Model I consists of 1082 species. In order to compute the series of diffusion flames that result in the extinction limit data of interest here, the resulting model must be reduced to a more manageable size. The reduction for flame conditions is performed by the path flux analysis technique [20] to produce a reduced high temperature model of 211 species. The requisite revalidation of this reduced high temperature model has been performed with appropriate modelling assumptions as summarised for example in Dooley et al. [21]. Of chief importance here is that computations with the combined reduced model show no significant loss of fidelity when compared to those of the parent models. It is shown in Fig. 2 that this condition is met and that the reduced model closely reproduces pertinent high temperature combustion kinetic measurements, in shock tube ignition delay [7,22] and laminar burning velocities [23,24]. It has not been possible to achieve truly grid independent solutions for laminar burning velocity computations with the detailed model (1082 species). However the detailed and reduced models compute very similar burning velocities for both fuels when resolved over similar numbers of grid points, thus indicating the fidelity of the reduced scheme.

2.1.2. Model II

To evaluate the impact of alternative descriptions of the small species chemistry on our analysis, a second kinetic model to describe nH/MB diffusion flames is also constructed. *Model II* is composed of the n-heptane submodel of Mehl et al. [25] which is a revision of the work of Curran and co-workers that is employed in Model I. The most important advance between these works is the incorporation of the recent small species chemistry derived from the works of Healy et al. [26], specifically cited to Johnson et al. [27] and Moc et al. [28]. As in Model I, the Dooley et al. methyl butanoate sub model is added to this construct. Model II consists of 702 species. As with Model I, a reduced scheme is required to compute the extinction limit, the path flux analysis technique produces a reduced Mode II of 242 species. The requisite validation of

detailed and reduced models is presented against the same pertinent high temperature combustion kinetic phenomena in Fig. 3.

Note that both reduced models share close fidelity to their respective detailed parent models and to experiment, Figs. 2 and 3. However, MB burning velocities are consistently computed to be higher than experiment, indicating an overestimation of flame reactivity that is consistent with the extinction limit comparisons discussed later. This behaviour is noted with two different small species chemistry subsets and points to the need for further improvements in the methyl ester oxidation scheme in order to reach true quantitative fidelity.

2.2. Transport model

CHEMKIN calculates the binary diffusion coefficients using an implementation of the kinetic theory of gases [16]. In terms of mass diffusion, the most important inputs required in this treatment are the Lennard-Jones parameters, of collisional diameter, σ , and the energy well depth, ϵ/k . In order to ascertain the uncertainty in these parameters, an intended-thorough literature review with an objective of evaluating the available parameters for oxygenated species has been performed.

An excellent review of the different techniques employed for the derivation of ϵ/k and σ and of their fidelities has been published by Mourits et al. [29], along with recommended values for ϵ/k and σ for a host of both polar and non-polar molecules including the esters, methyl acetate and ethyl acetate. Titiani et al. [30], Liu et al. [31], Mansoori et al. [32], Wang and Frenklach [33] and Holley et al. [34] have all employed various methods to derive ϵ/k and σ for a variety of molecules including the *n*-alkanes. The parameters presented by all these authors have been classified by functional group and presented in Figs. 4 and 5. Following Wang and Frenklach, analysis of the data for the collisional diameter parameter show a strong correlation to molecular weight, the similar relationship provided by these authors for aromatic species is also shown, and agrees well with the available experimental values. The available σ values for oxygenates show similar behaviour to aliphatic hydrocarbons, despite some obvious deviations. In fact once those molecules containing hetero-atoms (e.g. N,S,Cl) and the noble gases are removed from consideration, data for all molecules correlate to a large extent independently of molecular structure. A fit of the values presented for all aliphatic hydrocarbons, oxygenates and radical species produces the following equation:

$$\sigma = 1.468 M^{0.297} \quad (2)$$

Figure 5 shows that agreement between respective authors for a similar treatment of energy well depth is less cohesive. The majority of the presented ϵ/k parameters for oxygenates come from the review of Mourits et al. [29], and correlate poorly at low molecular weights, but appear to converge at higher molecular weights. Despite some recognised incongruities the limited available ϵ/k data for oxygenated species indicate that oxygenates show a similar correlative behaviour to molecular weight as that found for hydrocarbons, Fig. 5. An analogous treatment for collisional diameter, again ignoring hetero-atom species and noble gases, produces the following equation:

$$\epsilon/k = 41.68 M^{0.506} \quad (3)$$

The recommendations of Holley et al. [34] were also discounted as their data consistently occupy the upper range of values for the higher molecular weight species. The values presented by Wang and Frenklach for aromatic molecules show a similar bias in comparison to values derived from experiment, unsurprisingly so as both studies employ the same technique of corresponding states. Eqs. (2) and (3) have been used here to estimate all ϵ/k and σ values

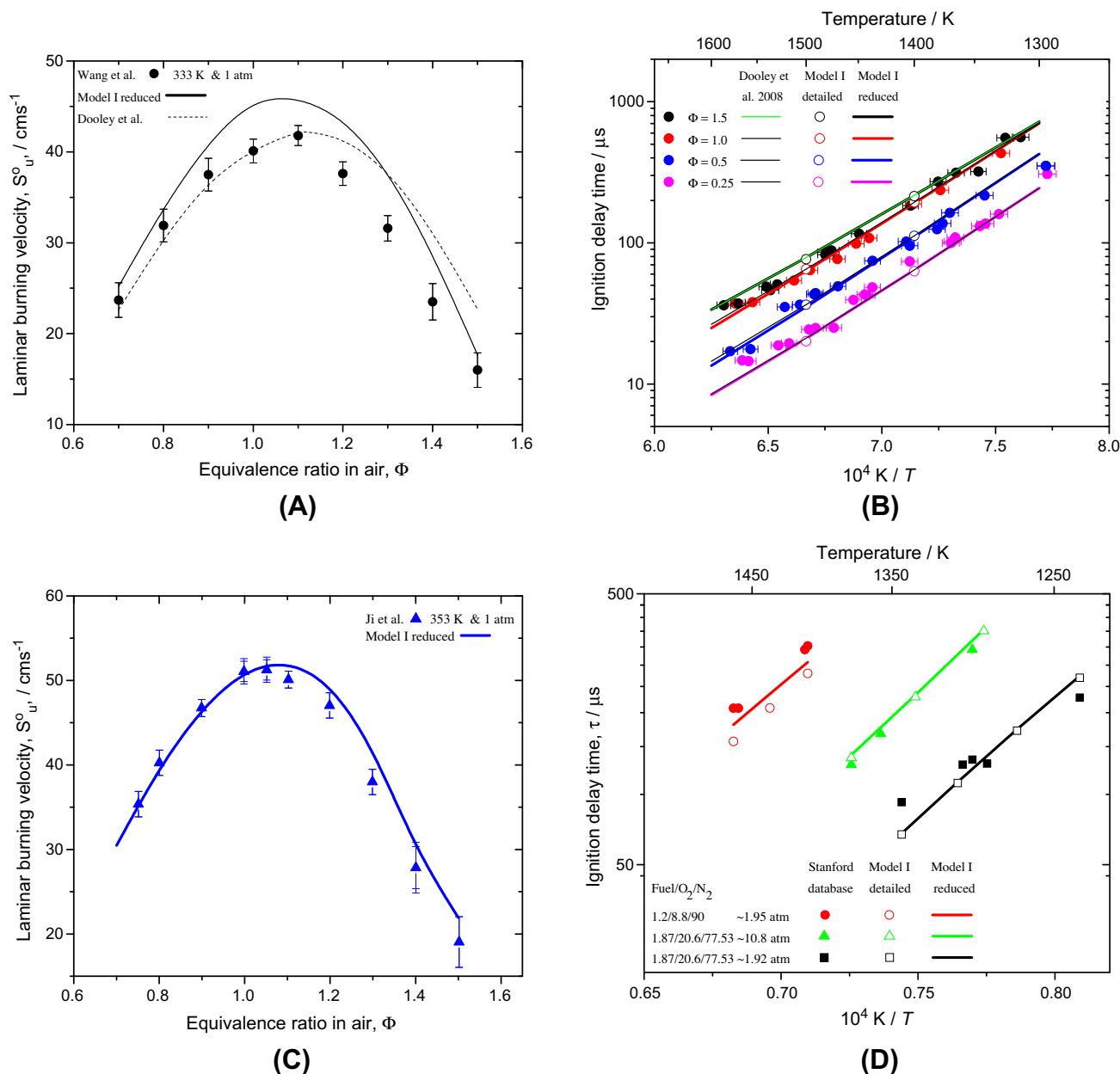


Fig. 2. Model I performance. Experiments (solid symbols), detailed model computations (hollow symbols) and reduced model computations (solid lines): (A) methyl butanoate laminar burning velocities at 1 atm and $T_i = 333 \text{ K}$ from Wang et al. [24], model computations of Dooley et al. [7] (dashed line), (B) methyl butanoate ignition delay times at 4 atm from Dooley et al. [7], model computations of Dooley et al. [7] (light lines), (C) n-heptane laminar burning velocities at 1 atm and $T_i = 353 \text{ K}$ from Ji et al. [23] and (D) n-heptane ignition delay times at select conditions from the Stanford data base [22].

for the species of the methyl butanoate sub model. The values for the n-heptane and small species sub models are adopted from the respective parent models. For n-heptane it is noted that σ and ε/k values generated by the assumption of corresponding states compare favourably to those produced by Eqs. (2) and (3); $6.25 \text{ \AA}/459 \text{ K}$ versus $5.77 \text{ \AA}/428 \text{ K}$. Based on the standard deviation of the above correlations, conservative uncertainties of $\pm 18\%$ in σ and $\pm 20\%$ in ε/k are estimated for the oxygenated species. The description of the transport parameters is completed as follows; the dipole moment of all species with the ester functionality is estimated as described by Daubert and Danner [35]. The Polarizability of ester type species is assigned in accordance with Hudson et al. [36], with the polarizable volume of MB assumed equal to that reported for methyl iso-butanoate. Finally, the rotational-collisional relaxation number is assumed to be unity for all species of the MB sub-model. These updated transport parameters have been utilised in the

simulation of all flames, unless otherwise stated. The impact of the uncertainty associated with these parameters on the computation of the extinction limit is discussed later, including an evaluation of the impact of parameters generated by the technique of corresponding states that employ critical temperature/critical pressure relations [37].

3. Results and discussion

The extinction strain rate, a_{E_s} , of MB is presented as a function of fuel mole fraction in Fig. 6. Dooley et al. [7] tested their work over a broad range of conditions involving premixed rather than environments dominated by diffusion. In this model the CHEMKIN transport description of MB as fuel was erroneously described as being atomic rather than molecular. While this is not of significant importance under premixed conditions, it is of controlling importance under

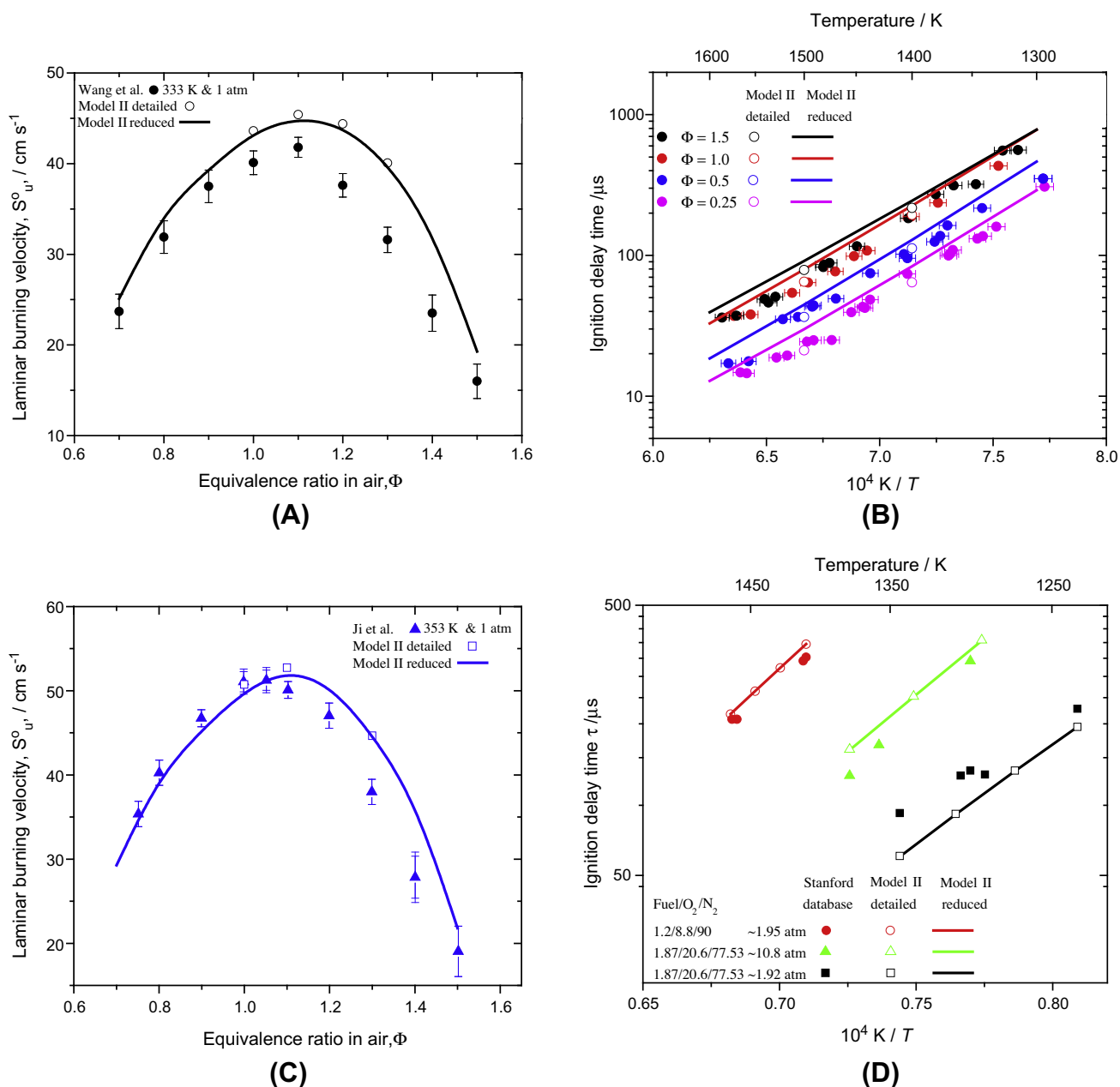


Fig. 3. Model II performance. Experiments (solid symbols), detailed model computations (hollow symbols) and reduced model computations (solid lines): (A) methyl butanoate laminar burning velocities at 1 atm and $T_i = 333$ K from Wang et al. [24], (B) methyl butanoate ignition delay times at 4 atm from Dooley et al. [7], (C) n-heptane laminar burning velocities at 1 atm and $T_i = 353$ K from Ji et al. [23] and (D) n-heptane ignition delay times at select conditions from the Stanford data base [22].

diffusion flame conditions. Hence employing the original transport description of Dooley et al. results in the over-prediction of the measured extinction strain rate by close to a factor-of-five, Fig. 6. When the erroneous transport description is replaced with that generated from the present work, the computed extinction strain rates are approximately a factor-of-two higher than the measurements. Finally, by adopting the small species chemistry of Curran et al. over that originally employed by Dooley et al. (Model I) the computed extinction strain rate deviates from experiment by ~ 30 –12%, depending on the MB fuel load.

3.1. Impact of transport parameters on kinetic modelling computations

For MB, we have estimated alternative values of $\sigma = 5.56$ Å and $\epsilon/k = 589$ K from the method of corresponding states as outlined

by Tee et al. [37]. These values are quite similar to the values generated by Eqs. (2) and (3), $\sigma = 5.80$ Å and $\epsilon/k = 433$ K. The disparity in σ value is within the uncertainties demonstrated in Fig. 4. Thus in the present study, the transport parameters for the species of the methyl ester submodel are produced by correlation to molecular weight, as the accuracy is comparable to the corresponding states approach, the method is relatively more simple (critical temperature, critical pressure and boiling data are not required), and the correlations are applicable to radical as well as stable species. In addition, this approach allows for meaningful uncertainties in σ and ϵ/k to be derived and their effects on flame computations evaluated. An exemplar test simulation was performed for the $X_{MB} = 0.11$ case with only the transport properties of MB adjusted to the apparent limits of certainty as fuel molecules are the only important high molecular weight diffusive species present in the flame [34]. The computed extinction strain rate with σ and ϵ/k values of 4.76 Å/519 K and

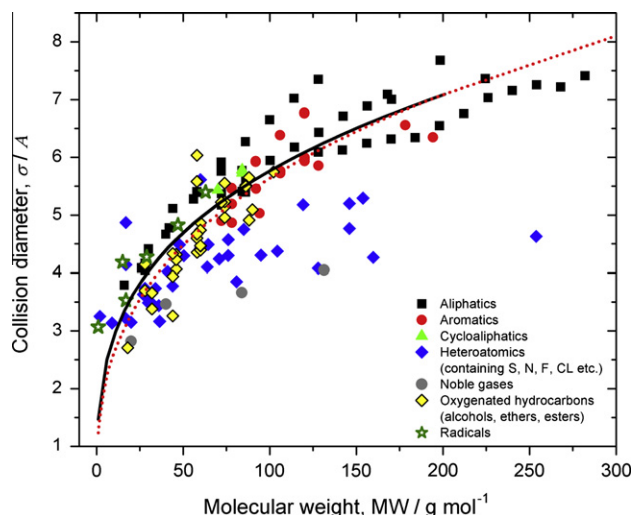


Fig. 4. Literature values of collision diameters versus molecular weight [29–37] (symbols), correlation for hydrocarbons and oxygenates from Eq. (2) (black line), and correlation for aromatics of [33] (red dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

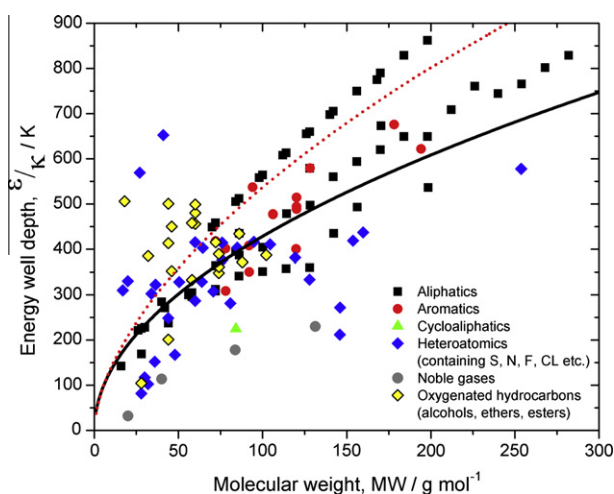


Fig. 5. Literature values of energy well depth versus molecular mass [29–37] (symbols), correlation for hydrocarbons and oxygenates from Eq. (2) (black line), and correlation for aromatics of [33] (red dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6.84 Å/346 K is 156/215 s^{−1} respectively. Thus from the baseline value of 180 s^{−1}, an approximate uncertainty in the computed extinction strain rate of 13–19% due to the apparent worst case uncertainty in the estimation of Lennard–Jones parameters is produced. Such is represented by exemplar uncertainty bars in Fig. 7.

3.2. n-Heptane/methyl butanoate diffusion flame extinction

Normal-heptane, which is chemically representative of petroleum and Fischer–Tropsch derived fuels, is observed to have extinction strain rates that are approximately three times higher than measured for MB when compared on a molar basis, Fig. 7. Additionally, the extinction strain rate of MB shows a linear dependence on fuel mole fraction, that of the alkane fuel is distinctly non-linear. To explore and better parameterize these effects as they relate to biodiesel fuels, two blends of nH/MB were investigated (73/27,

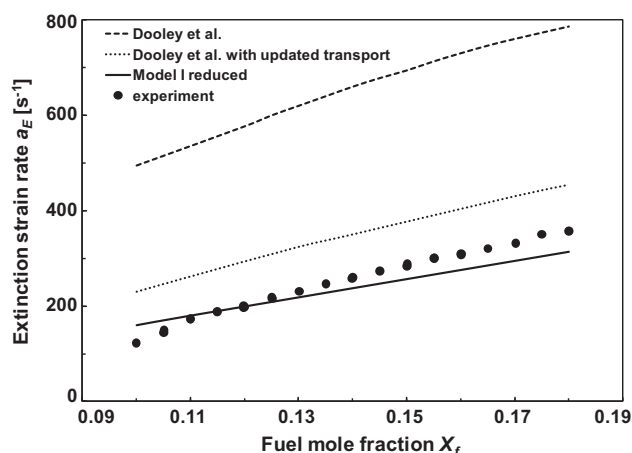


Fig. 6. Methyl butanoate extinction strain rates, experiment (symbols) and model computations; Dooley et al. [7] (dashed lines), Dooley et al. with transport update (dotted lines, see text) and Model I reduced (solid line).

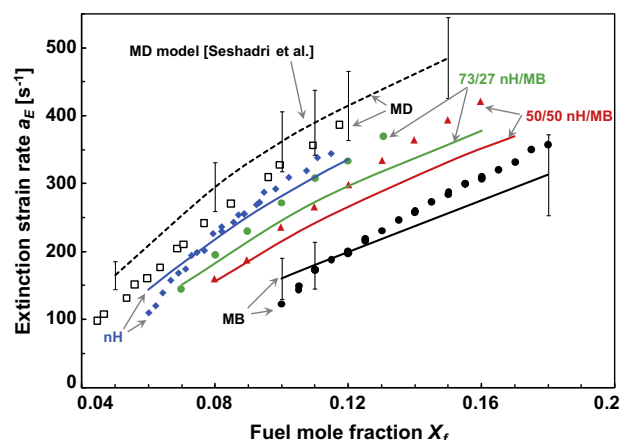


Fig. 7. Extinction strain rates of methyl butanoate (MB), n-heptane (nH), nH/MB mole% and methyl decanoate (MD) diffusion flames. Experiment (symbols) and computations (lines). Solid lines are Model I reduced, dashed line is Seshadri et al. [38]. Exemplar error bars represent uncertainties affected by apparent worst case uncertainty in molecular transport parameters, see text.

and 50/50 mol% mixtures). The results show that the addition of MB to nH results in a systematic reduction in flame reactivity as measurable by a decreasing extinction strain rate, Fig. 7. These findings imply flame stability variations for biodiesel/petroleum fuel blends and may indicate significant fundamental differences in the respective oxidation mechanisms of alkane and methyl ester fuels at flame conditions.

3.3. Methyl decanoate diffusion flame extinction

Actual biodiesel components possess a much larger alkyl chain than MB and so the extinction behaviour of such molecules ought to have combined character of both nH (n-alkyl) and MB (methyl ester). To examine this hypothesis, the extinction behaviour of the larger biodiesel surrogate, methyl decanoate, MD, has been measured and also shown in Fig. 7. The extinction strain rate of MD on a molar basis is observed to be slightly higher than that of nH. This is somewhat surprising, given that the calorific value of MD is much larger than that of nH ($\Delta H_{\text{combustion}}(\text{MD})$ – 1648 kcal mol^{−1} vs $\Delta H_{\text{combustion}}(\text{nH})$ – 1159 kcal mol^{−1}). MD extinction limit computations have been conducted with the kinetic model

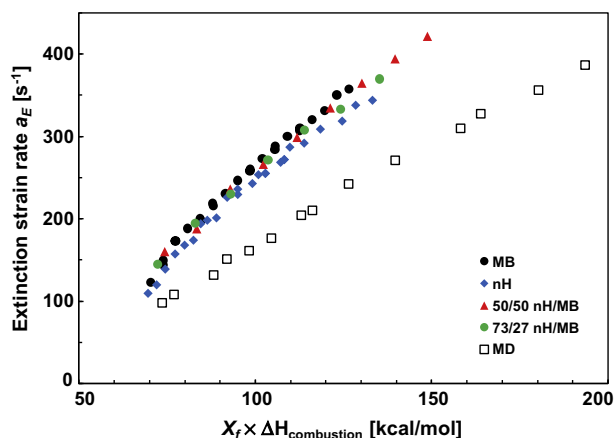


Fig. 8. Extinction strain rates of methyl butanoate (MB), n-heptane (nH), nH/MB and methyl decanoate (MD) diffusion flames versus normalised energy content, the product of fuel mole fraction and standard state enthalpy of combustion.

constructed by Herbinet et al. [11] and reduced and tested by Seshadri et al. [38], also in Fig. 7. The Seshadri et al. MD model over-estimates the extinction data, failing to capture the inhibitive effect of the ester moiety sufficiently. MD transport parameters of $\sigma = 7.30 \text{ \AA}$ and $\varepsilon/k = 604 \text{ K}$ as suggested by Seshadri et al. (from Herbinet et al.) and are comparable to values of $\sigma = 6.93 \text{ \AA}$ and $\varepsilon/k = 586 \text{ K}$ that result from our literature review. For the $X_{MD} = 0.11$ case an extinction strain rate of 384 s^{-1} is computed with the transport description of Seshadri et al. The equivalent computations using the transport parameters produced by Eqs. (2) and (3) results in just a 1% increased extinction strain rate of 387.5 s^{-1} . Extinction limits computed with the L–J parameters owing to the maximum and minimum uncertainties derived from Figs. 4 and 5 are displayed as error bars in Fig. 7 and equate to an uncertainty in the computed extinction limit of the order of 7–12%. In terms of kinetic model development, diffusive parameter uncertainties are significant. If Lennard–Jones parameters of the upper bounds of certainty are applied, they may be sufficient to reconcile the lower flame reactivity of MD experiment relative to the model of Seshadri et al.

Figure 8 compares the extinction strain rates for all fuels to the normalised energy content produced by taking the sum of the standard state enthalpy of combustion and fuel mole fraction. As the mass transport parameters of nH and MB are both approximately equal (very similar molecular weights) this metric regulates the (peak) flame temperature. Thus the relative chemical kinetic contribution to the extinction strain rate may be more fairly assessed. The extinction behaviour of nH and MB are very similar by this comparison, indicating that the inhibitive effect of MB blending in nH diffusion flames is mainly due to the lower energy content of MB relative to nH at enthalpies of combustion of $703.3 \text{ kcal mol}^{-1}$ and $-1159.6 \text{ kcal mol}^{-1}$ respectively, Fig. 8.

The extinction behaviour of MD is quite striking by this comparison, demonstrated to be almost a factor of two lower than that of n-heptane. Without the interpretation allowed by a validated kinetic model it is presently not possible to separate the contributing chemical kinetic and molecular transport effects to the extinction strain rate [12,13]. On a molar basis, one can expect an equivalent inhibitive effect as observed for MB relative to nH, to be present for larger methyl esters relative to equivalent alkanes.

3.4. Kinetic modelling analysis of n-heptane and methyl butanoate diffusion flame extinction

The extinction limits of hydrocarbon diffusion flames in counter flowing systems are known to be governed by a balance between

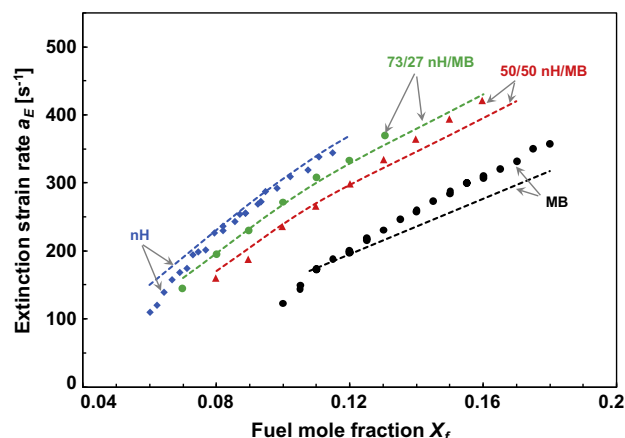


Fig. 9. Model II computations of extinction strain rates of methyl butanoate (MB), n-heptane (nH) and nH/MB mole% diffusion flames. Experiment (symbols) and Model II reduced computations (dashed-lines).

heat release in the reaction zone and heat losses to fuel and oxidizer sides owing to fuel consumption and convective heat transfer to the cold bath gases [15,39,40]. Additionally, fuel fragment chemistry is centrally involved through the formation of the radical pool, especially the hydroxyl (OH) radical, the reactions of which contribute heavily to heat release through water and carbon dioxide formation, sustaining the flame [12,13]. Both Model I and Model II generally reproduce the experimental extinction strain rate data well, Figs. 7 and 9 respectively, with the relative inhibitive effects of the methyl ester qualitatively reproduced. However both modeling incarnations show relatively poorer agreement for the methyl ester versus the alkane, indicating that improvements in the prescribed methyl ester chemical kinetics and/or thermochemical parameters, or quite possibly even in the small species chemistry more specifically pertinent to the oxidation of such oxygenates are required.

A reasoning as to why the methyl ester functionality has a distinctive effect on diffusion flame chemical reactivity, and of what the origin of the deficiencies in the presented MB and MD models may be can be garnered from a detailed analysis of the nH/MB model computations. Both Model I and Model II exhibit similar fidelity to experiment, thus the coming section presents a detailed analysis of the Model I computations. However, though not discussed in detail, Model II computations have been identically analysed and representations of this analysis are also provided. Despite some expected numerical differences due to the different small species descriptions employed, it is important that both models provide the same conclusions on diffusion flame extinction, and on the chemical mechanism that contributes to extinction specific to methyl esters over n-alkanes.

Figure 10 explains in a summarised schematic form, the general chemical kinetic mechanism of nH and MB diffusion flame extinction, the parametric explanation of which follows in detail. Briefly, vinyl and formyl radicals are the major mechanistically important (reactive) intermediate species formed in these flames. Both predominantly decompose to release H atoms at low strain rates. Hydrogen atoms sustain the flame by participating in radical chain branching by reaction with O_2 . At higher strain rates, an increased mixing of these reactive intermediates with O_2 mainly due to the thinned flame width is affected, resulting in the formation of hydroperoxy radicals instead of H atoms. The hydroperoxy radical inhibits the flame by reaction with H atoms and OH radicals, which may otherwise be involved in the coupled dominant radical chain branching and exothermic flame reactions of $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ and $\text{OH} + \text{CO} = \text{CO}_2 + \text{H}$ respectively.

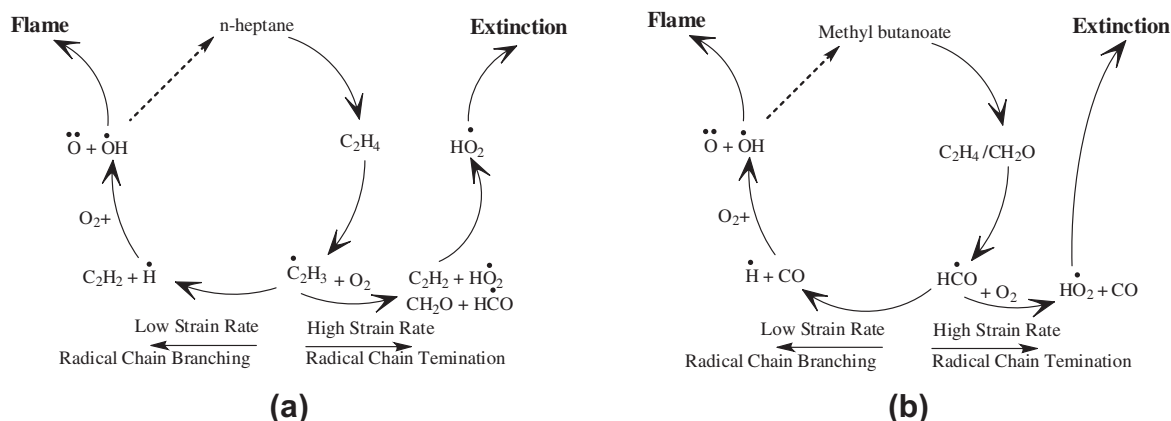


Fig. 10. Conceptual schematic of the contribution of molecular transport/chemical kinetic coupled mechanism to the extinction of (a) n-heptane and (b) methyl butanoate laminar diffusion flames: hydrogen atom production sustains the flames and hydroperoxy radical production inhibits both flames, see text. Ethylene (C_2H_4) consumption mechanism is common to both flames.

These conclusions are explained by the analysis of representative $X_{fuel} = 0.11$ flames for (1) nH, (2) MB and (3) 50/50 mol% nH/MB at conditions of low strain rate and very close to the extinction limit. The role of each elementary reaction of the kinetic model in defining the important flame properties has been analysed by integrating over the entire reacting domain:

3.4.1. Heat release rate

Figure 11a shows the maximum heat release rate and flame temperature versus strain rate. The heat release rate and flame temperature of MB diffusion flames is lower than those of nH diffusion flames. This may simply be related to the lower energy content of MB compared to nH as apparent by standard state

enthalpies of combustion of $-703.3 \text{ kcal mol}^{-1}$ and $-1159.6 \text{ kcal mol}^{-1}$ respectively. However, a difference in energy content of 40% is not sufficient to explain completely the chemical kinetic differences in extinction behaviour.

For all fuels, the model shows that the heat release rate increases with increasing strain rate whilst the maximum flame temperature is shown to drop. As described by Sung et al. [15], the heat release rate is linearly correlated with fuel flux into the reaction zone, which is also a linear function of strain rate. As the strain rate is increased in a counterflow flame the flame thickness is reduced, leading to an increased rate of heat loss to the fuel and oxidizer sides as the temperature profile becomes more abrupt. Accordingly, as strain rate is increased, the increase in heat release rate

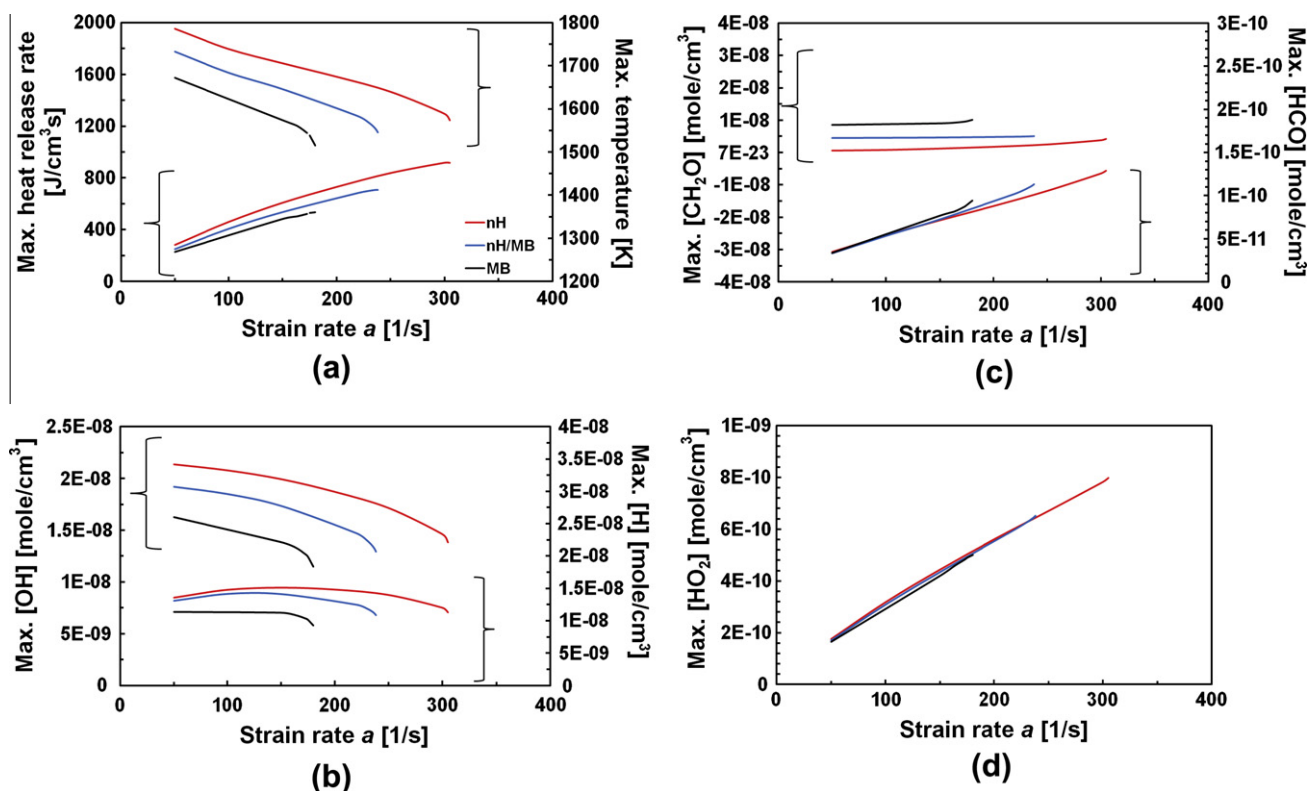


Fig. 11. Maximum values from Model I computations of n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames versus strain rate; (a) heat release rate and flame temperature, (b) OH radical and H atom concentrations, (c) CH_2O and HCO radical concentrations and (d) HO_2 radical concentrations for $X_{fuel} = 0.11$.

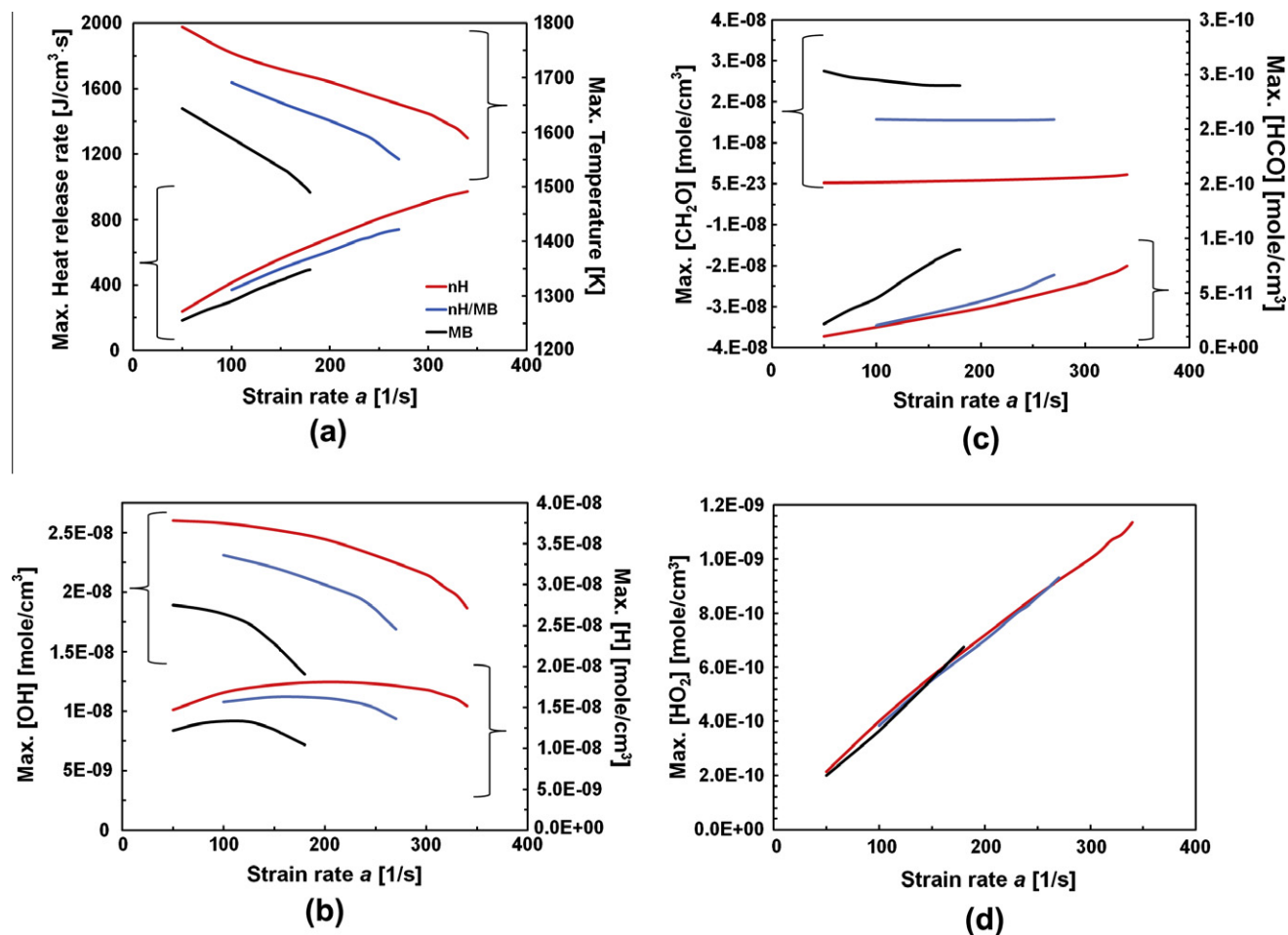


Fig. 12. Maximum values from Model II computations of n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames versus strain rate; (a) heat release rate and flame temperature, (b) OH radical and H atom concentrations, (c) CH₂O and HCO radical concentrations and (d) HO₂ radical concentrations for $X_{fuel} = 0.11$.

is offset by larger heat losses, resulting in a decrease in flame temperature as is observed in the flame simulations with both models, Figs. 11 and 12.

3.4.2. The role of the hydroxyl, OH, radical

Figure 11b shows the dependence of the peak concentrations of the dominant flame radicals, OH and H, on strain rate. From nH to nH/MB to MB diffusion flames, there is a stepwise decrease in both. For all flames, the OH concentration was found to decrease monotonically with increasing strain rate. Model I computations show that the dominant high temperature chain branching reaction, $H + O_2 = O + OH$, accounts for almost 50% of total OH radical formation for all flames at all strain rates. The positive temperature dependence of this reaction and the lower flame temperatures experienced at high strain rates contribute to the computed decrease in OH radical concentration with increased strain rate.

The OH radical population has a controlling influence on flame reactivity through its role in two principle elementary reactions. For all fuels and all strain rates studied, the OH radical is consumed most significantly by the reactions of $OH + CO = CO_2 + H$ and $OH + H_2 = H_2O + H$ which in combination, are responsible for consuming more than half of the total OH produced. These reactions are each doubly active in promoting flame reactivity. Not only are both reactions exothermic (the third and fourth largest producers of heat), but each also converts an OH radical to a H atom that in turn result in chain branching through $H + O_2 = O + OH$. The occurrence and coupling of these three reactions strongly influence

flame reactivity. Thus, the differences in model performance in computing the flame reactivity of nH, MB and nH/MB diffusion flames is indicative of significantly different chemical pathways that support the population of the OH radical within the different flames.

3.4.3. Fuel consumption

Both fuels are predominantly decomposed by H abstraction/alkyl radical beta-scission reactions. Hydrogen atom is the principle abstractor, consuming 85% and 92% of fuel respectively in nH and MB single component flames. Although the generic mechanism of fuel consumption is similar for nH and MB, it is noteworthy that the intermediate species formed differ considerably and are specific to the respective fuel chemistries.

3.4.4. Formaldehyde formation

Figure 13a shows the spatial distribution of CH₂O and OH radical for nH and MB diffusion flames near the extinction limit. Compared to nH diffusion flames, MB diffusion flames form significantly larger amounts of CH₂O, up to a factor-of-five higher in peak value relative to nH flames. Figure 13b compares the fuel consumption rates in each flame relative to the production rate of CH₂O. Formaldehyde results directly from the H abstraction/beta-scission of the MB fuel structure, whereas its production in nH flames results from the small species reactions of the main body of the reaction zone, somewhat after nH fuel has been consumed.

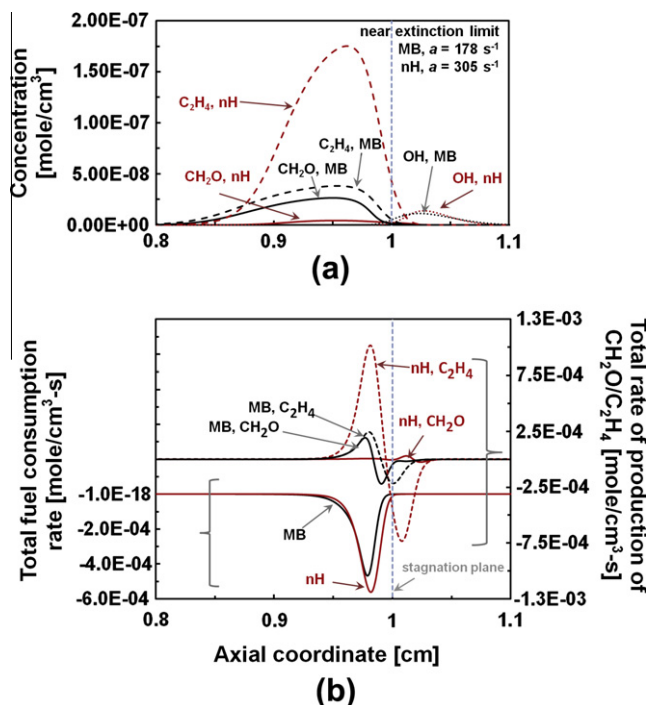


Fig. 13. Model I computations for select species in n-heptane (nH) and methyl butanoate (MB) diffusion flames close to extinction for $X_{fuel} = 0.11$, species concentrations (a) and total rates of production (b).

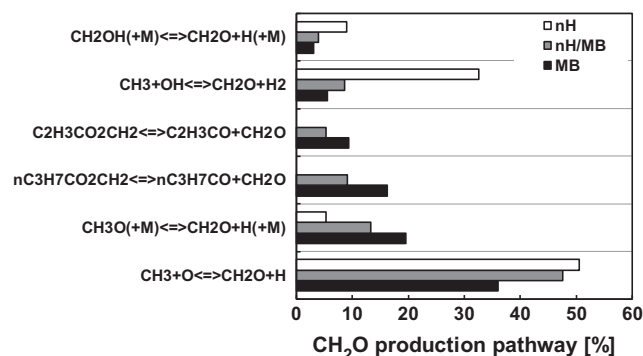


Fig. 14. Model I computations of CH_2O production pathways for n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames close to extinction for $X_{fuel} = 0.11$.

This fundamental difference in flame structure can be more clearly understood by considering the production pathways of CH_2O which are presented in Fig. 14. For nH diffusion flames, the dominant CH_2O production pathways are through the oxidation of the CH_3 (methyl) radical via O atom and OH radical, accounting for approximately 80% of total CH_2O production. These main body reaction zone processes account for the spatial location of the peak CH_2O production rate in Fig. 13b. For MB diffusion flames, significant CH_2O production comes from the decomposition of MB radical fragments, which locate the peak CH_2O concentration at a much earlier point in the flame structure. Depicted in Fig. 16, the beta-scission of the methyl ester radicals ($nC_3H_7CO_2CH_2$ and $C_2H_3CO_2CH_2$) release formaldehyde directly and account for a combined 26% of total CH_2O produced in the MB flame. A further 20% is produced by the decomposition of the methoxy radical, which is predominantly formed (80% of total) by reactions of the MB sub mechanism. These pathways are specific to methyl ester oxidation

and will also occur for real biodiesels. Note that analysis of Model II computations shows the same main features, Figs. 12 and 15.

3.4.5. Sensitivity analysis

A sensitivity analyses of each elementary reaction to extinction strain is useful in interpreting the controlling factors to diffusion flame extinction of a kinetic nature. Sensitivity coefficients have been generated on a logarithmic basis by perturbing the pre-exponential factor of each individual rate constant of Model I and observing the effect of the adjustment on the computed extinction strain rate. A positive sensitivity coefficient indicates that an increase in reaction rate constant promotes flame reactivity, a negative sensitivity coefficient indicates an inhibitive effect or weakening of the flame. This sensitivity analysis is presented in Fig. 17 and shows the respective fuel chemistries to be of minor importance. Though some ester-specific reactions show modest sensitivity, e.g. $MB + H$ and CH_3OCO decomposition, adjusting the rate constants of these reactions, to any reasonable extent of uncertainty, cannot effect computed extinction limits at all strongly.

However, the sensitivity results do elucidate the importance of the formation and consumption of the HO_2 radical on extinction behaviour. Interestingly the impact of the HO_2 chemistry set, which is common for each fuel, differs substantially between each fuel. In MB diffusion flames, the sensitivity analysis also implies the significance of CH_2O formation and consumption through the HCO (formyl) radical, Fig. 17. Analysis of the chemical flux shows strong differences in the extent to which the oxidation of CH_2O proceeds through the $HCO \Rightarrow H + CO$ or the $HCO + O_2 \Rightarrow HO_2 + CO$ pathway in each flame, such is evident by the peak concentrations of these species shown in Fig. 11c. The decomposition of the formyl radical releases reactive H atoms that can lead to radical chain branching. Conversely, if this radical reacts bimolecularly, carbon monoxide and a stable species, or in the case of reaction with O_2 , HO_2 is formed. The consequences of this duplicity for diffusion flame reactivity is explained below.

3.4.6. The role of the hydroperoxy, HO_2 , radical

Both Models I and II show the HO_2 radical concentration to increase with strain rate for all flames until extinction is observed, Figs. 11d and 12d respectively. Analysis of the Model I computations shows that the formation of the HO_2 radical strongly influences flame reactivity. At conditions close to extinction, >90% of this radical is consumed by reaction with H, O, and OH radicals. Though perhaps not immediately evident, these processes all inhibit flame reactivity. This may not be apparent without careful consideration of the sensitivity analysis which apparently indicates that the reaction of $HO_2 + H = OH + OH$ promotes flame reactivity, Fig. 17.

There is close competition for $HO_2 + H$ reactants between product sets of $OH + OH$ and $H_2 + O_2$, respectively accounting for 30–39% and 6–8% of total HO_2 consumption. The formation of the chain propagating $OH + OH$ products set enhances flame reactivity only relative to the alternative $H_2 + O_2$ exothermic but markedly chain terminating product set. Hence the observed sensitivity coefficients. However it is crucial to realise that any reaction of $HO_2 + H$ inhibits flame reactivity by interfering with alternative reactivity-enhancing channels to H atom consumption. HO_2 formation in the reaction zone, where H atom concentrations are highest, directly mitigates the extent of the chain branching reaction, $H + O_2 = O + OH$, which is the ultimate reactivity enhancing reaction under flame conditions. The sensitivity analyses demonstrates this.

The second major HO_2 consumption route (25–38% of total) is through $HO_2 + OH = H_2O + O_2$. This reaction is the fourth largest single contributor to heat liberation and supports the flame through its exothermicity. However, it has a duplicity in affect as it inhibits the flame by consuming two radicals. Whilst there is a

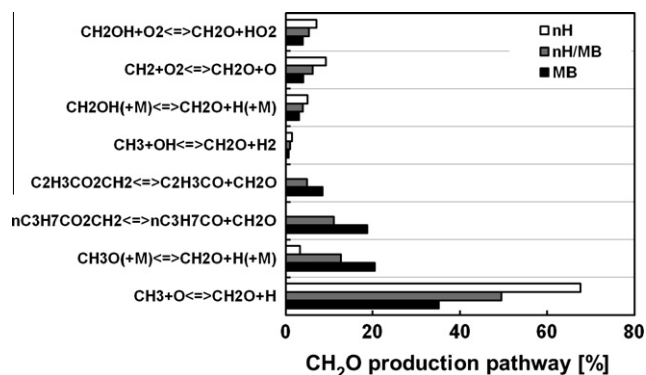


Fig. 15. Model II computations of CH₂O production for n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames close to extinction for $X_{fuel} = 0.11$.

balance between the self contained net influence of this reaction, it is not comparable to the overall inhibition it causes through the consumption of OH radicals. The dominant consumption process of the OH radical is the exothermic (and hydrogen atom producing) reaction of $OH + CO = CO_2 + H$. The production of increased quantities of HO₂ inhibits and eventually quenches the flame. The chemistry that describes the elementary processes of these small radicals is common to the reacting systems of both fuels. Therefore, to understand the difference in the oxidation chemistry of the respective fuels, a flux analyses of the reactions responsible for HO₂ production in each flame is presented in Fig. 18.

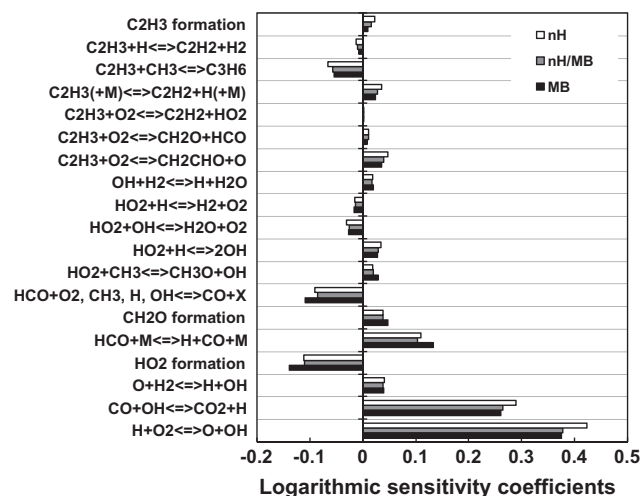


Fig. 17. Sensitivity coefficients for the reactions of Model I to the extinction strain rate of n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames for $X_{fuel} = 0.11$.

3.4.7. Hydroperoxy, HO₂, radical production

Shown in Fig. 18 by Model I and by Model II in Fig. 19, for all flames, the main mode of HO₂ production is $H + O_2 + M = HO_2 + M$, accounting for approximately 85% of total HO₂ formed for each flame at low strain rates but for considerably less close to extinction. As these flames approach the extinction condition, there is a

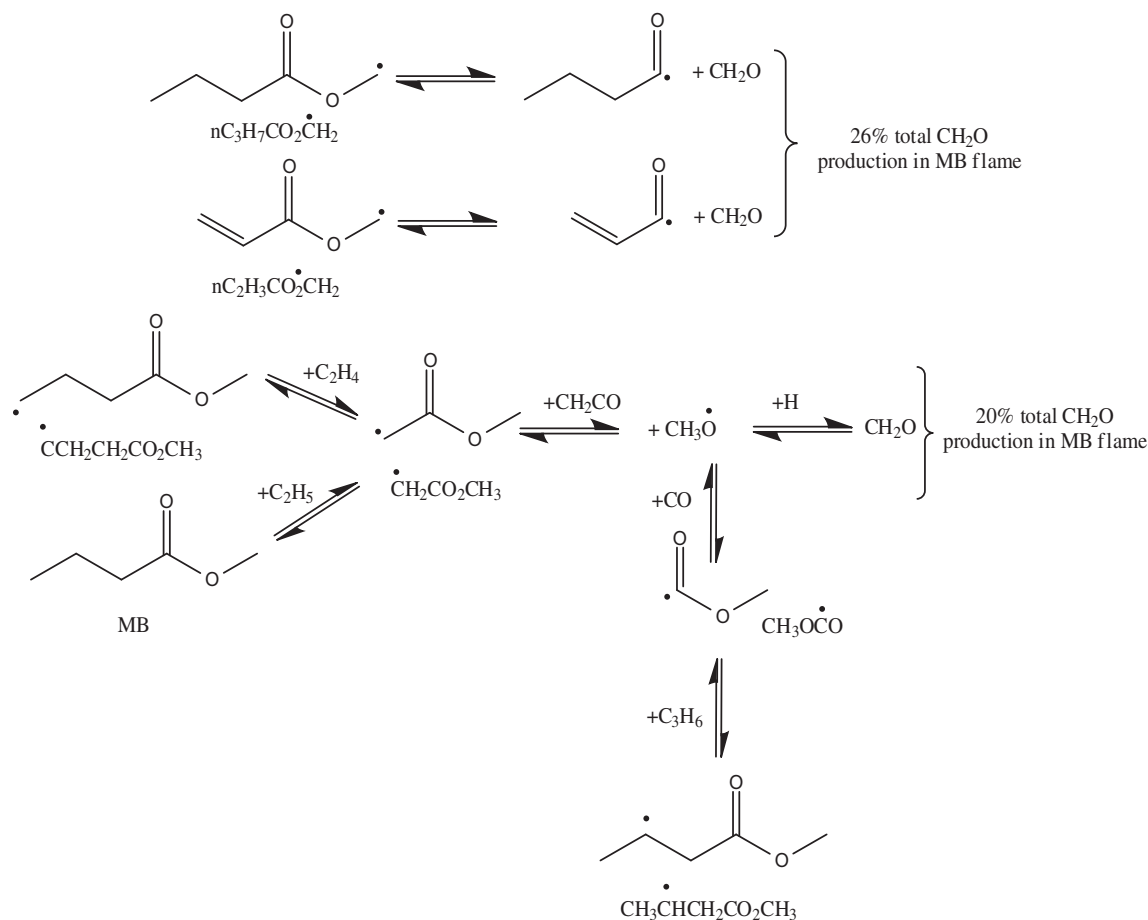


Fig. 16. Model I computations for CH₂O production pathways from the fuel chemistry for a methyl butanoate (MB) diffusion flame close to extinction for $X_{fuel} = 0.11$.

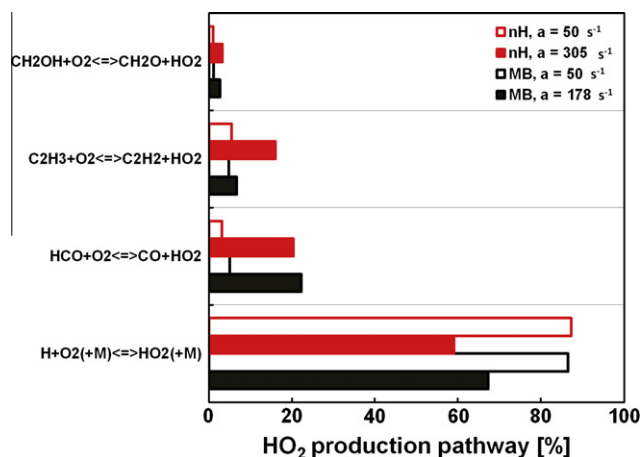


Fig. 18. Model I computations of HO₂ production pathways for n-heptane (nH), methyl butanoate (MB) and 50/50 mol% nH/MB diffusion flames at low strain rates and close to extinction for $X_{fuel} = 0.11$.

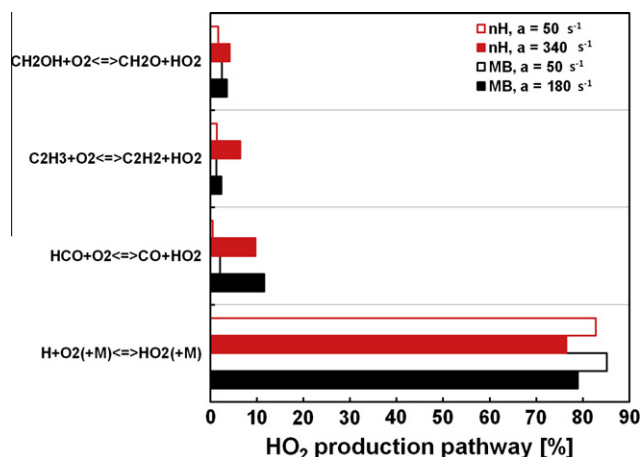


Fig. 19. Model II computations of HO₂ production for n-heptane (nH) and methyl butanoate (MB) diffusion flames at low strain rates and close to extinction for $X_{fuel} = 0.11$.

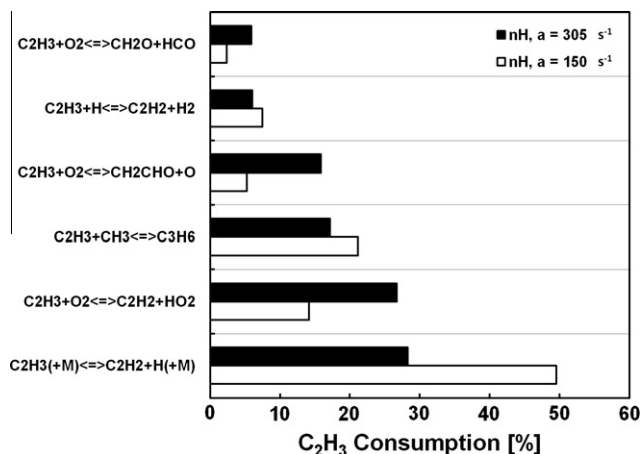


Fig. 20. Model I computations of vinyl (C₂H₃) radical consumption for an n-heptane diffusion flame at low strain rate (150 s⁻¹) and close to extinction (305 s⁻¹) for $X_{fuel} = 0.11$.

The chemical flux analyses show important differences in fuel fragment contributions to HO₂ formation in each of the differently fueled flames. When MB is the fuel, HO₂ is produced by formyl radical reaction with oxygen, $HCO + O_2 = CO + HO_2$. If nH is the fuel, HO₂ results from reaction of O₂ with the vinyl (C₂H₃) radical, $C_2H_3 + O_2 = C_2H_2 + HO_2$. Note that the same essential features are exhibited by both assembled models.

3.4.8. The role of the vinyl, C₂H₃, radical

The vinyl radical is produced via hydrogen abstraction from ethylene (C₂H₄) by the generic reaction path; $nC_7H_{15} \Rightarrow (nC_5H_{11} +) C_2H_4 \Rightarrow C_2H_3$. Ethylene is the most prominent intermediate in the oxidation of n-heptane, which unlike MB exhibits an extreme bias to the production of C₂H₄ over CH₂O, Fig. 13.

The consumption of the vinyl radical is complex, but may be divided into chain propagating/branching and chain terminating reaction classes. Sensitivity analysis shows that in general, C₂H₃ radical production exerts a promoting effect on flame reactivity. The consumption pathways of the vinyl radical are presented in Fig. 20 for pure nH diffusion flames. The vinyl radical has two significant processes which terminate the radical chain, $C_2H_3 + CH_3 = C_3H_6$ and $C_2H_3 + H = C_2H_2 + H_2$. All other vinyl radical reactions show positive sensitivity coefficients as a result of their direct competition with these inhibitive reactions, Fig. 17.

The reaction $C_2H_3 + O_2 = C_2H_2 + HO_2$, exerts only a weak positive sensitivity on the extinction strain rate. The alternative vinyl radical consumption routes, $C_2H_3 + O_2 = CH_2CHO + O$, $C_2H_3 + O_2 = CH_2O + HCO$ and $C_2H_3 (+M) = C_2H_2 + H (+M)$ all much more significantly enhance flame reactivity (due to their potential chain branching character which is either realised directly or eventually through the eventual production of H atoms). Therefore in absolute terms, $C_2H_3 + O_2 = C_2H_2 + HO_2$ is inhibitive to diffusion flame reactivity. This concept becomes important when oxygen concentrations are high, as the rate of $C_2H_3 + O_2$ is increased, as occurs at high strain rates when the reaction zone becomes constricted.

As strain rate is increased to the point of extinction, examination of the chemical flux through the C₂H₃ radical in an nH diffusion flame, Fig. 20, shows increased competition between direct H atom production through (unimolecular) decomposition and the overall less reactive bi-molecular reactions. Particular to this point is the increased contribution to consumption of the vinyl radical by $C_2H_3 + O_2$ resulting from the constriction of the reaction zone due to the high strain rate [39,40] and the consequent increased mixing of fuel/fuel fragments with oxidizer.

The increased rate of $C_2H_3 + O_2$ reactions is also apparent from Fig. 18. The formation of C₂H₂ + HO₂ is competitive with C₂H₃ decomposition and thus contributes to the increased levels of HO₂ with increased strain rate shown in Fig. 11d. This formation of HO₂ causes the explained inhibition of the small species processes which eventually in communication with physical effects [15,39,40] leads to extinction.

3.4.9. Evaluation of hydroperoxy, HO₂, radical, production in methyl ester flames

Hydroperoxy radical production in methyl butanoate flames is less dependent on the $C_2H_4 \Rightarrow C_2H_3 \Rightarrow$ products pathway, as in comparison to nH flames the ester molecular structure does not produce as significant quantities of ethylene upon oxidation. Ethylene is produced primarily through $CH_2CH_2CH_2CO_2CH_3 \Rightarrow C_2H_4 + CH_2CO_2CH_3$, which is accompanied by the eventual formation of an approximately equivalent amount of CH₂O, as depicted by Fig. 13.

In MB flames, vinyl radical formation/consumption is replaced or augmented with a more significant contribution from the formation/consumption of the formyl radical through formaldehyde oxidation. Whilst there is intense competition for the $C_2H_3 + O_2$ reactants between products of $CH_2CHO + O$, $CH_2O + HCO$ and

larger percentage contribution to HO₂ formation from fuel fragment reactions.

$C_2H_2 + HO_2$, all channels show very modest temperature dependence at flame conditions. There is no similar diversity of formyl radical consumption. Carbon monoxide and HO_2 are the only product set available to $HCO + O_2$ reactants. Thus high levels of HCO induce flame inhibition at the better mixed fuel/oxidizer conditions of high strain rates.

To test these conclusion some simple numerical experiments have been conducted by manipulating the chemical kinetic model and recomputing the extinction strain rate:

- (1) The methyl ester chemistry has been re-written so as to divert the chemical flux away from formaldehyde formation by allowing alternative products to be formed. For example, the $nC_3H_7CO_2CH_3$ radical is principle to CH_2O production in the MB flame. Without adjusting the Arrhenius parameters for the forward reactions, this radical is hypothesised to decompose to form C_2H_4 and the $CH_2CO_2CH_3$ radical which is hypothesised to decompose to CO_2 and the C_2H_5 (ethyl) radical (rather than to CH_2CO (ketene) and the CH_3O (methoxy) radical). This hypothetical kinetic description computes extinction strain rates which are $10\text{--}18\text{ s}^{-1}$ higher than those of Model I across $X_{MB} = 0.11\text{--}0.18$, i.e. 7–8% higher, confirming the inhibitive effect of CH_2O formation.
- (2) The extent to which the product set multiplicity of $C_2H_3 + O_2$ chemistry affects n-heptane diffusion flame extinction is tested by hypothesising that vinyl radical consumption occurs analogously to the consumption of the formyl radical, i.e. the $CH_2CHO + O/CH_2O + HCO$ product sets are not considered and the reaction of $C_2H_3 + O_2 = C_2H_2 + HO_2$ is allowed to occur with the same forward rate constant as the reaction of $HCO + O_2 = CO + HO_2$. This alternative kinetic model computes extinction strain rates which are $20\text{--}24\text{ s}^{-1}$ lower than those computed by Model I across $X_{nH} = 0.11\text{--}0.18$, i.e. 8–14%, confirming the inhibitive effect of $C_2H_3 + O_2 = C_2H_2 + HO_2$.

These numerical experiments provide somewhat crude quantifications of the chemical mechanism contributing to diffusion flame extinction. The chemical kinetic analysis and conclusions made on the nH/MB reacting system can be directly extended to much larger and more biodiesel like methyl esters, such as methyl decanoate. An improved quantitative understanding of formaldehyde production from methyl ester specific reactions is necessary for improved kinetic model performance for methyl ester fuels under flame conditions.

4. Conclusions

Using the diffusion flame extinction strain rate as a diagnostic measure, the diffusion flame reactivity of methyl butanoate, is observed to be approximately two times less than n-heptane when compared on a molar basis. The extinction strain rate of n-heptane/methyl butanoate diffusion flames declines significantly as methyl butanoate is added to n-heptane.

The available transport parameters for oxygenated species have been reviewed and are found to be not significantly different to those of similar molecular weight hydrocarbons. The apparent uncertainties in the estimation of the Lennard–Jones parameters leads to an uncertainty in computed extinction strain rate of the order of 10%. Consequently diffusion flames should not be used for initial development of the kinetic and chemical pathway components of combustion models for fuel systems that have a large uncertainty in diffusive parameters. These environments are much more appropriate for examination of transport elements once the accuracy of the kinetic and chemical pathway components has been determined under fully premixed conditions.

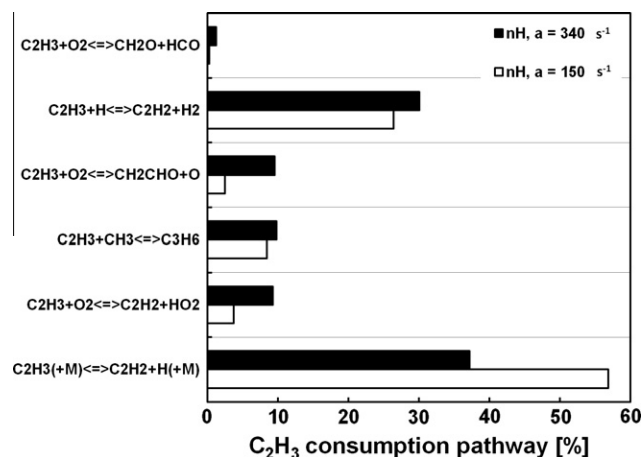


Fig. 21. Model II computations of vinyl (C_2H_3) radical consumption for n-heptane diffusion flames at low strain rate (150 s^{-1}) and close to extinction (340 s^{-1}) for $X_{fuel} = 0.11$.

Kinetic models to describe the diffusion flame reactivity of n-heptane/methyl butanoate diffusion flames are assembled, tested and reduced in order to fundamentally interpret the extinction limit observations. Two separate kinetic model assemblies are found to reproduce the experimental measurements quite well when appropriate transport parameters are incorporated.

As the molecular transport parameters of both n-heptane and methyl butanoate are very similar, analysis of the modelling computations for both pure and binary fuel flames show that the differences in diffusion flame extinction behaviour is related to the energy content of the fuel.

The extinction strain rate of each fuel is suggested to be dependent on the chemical kinetic potential of each fuel to form the hydroperoxy radical. The effect of hydroperoxy radical formation is identical for each fuel, contributing to flame extinction through its reactions with the small species population, particularly through the reactions of $H + HO_2 = \text{products}$ and $HO_2 + OH = H_2O + O_2$. The occurrence of these reactions have the effect, both directly and indirectly, to reduce the chemical flux through the $H + O_2 = O + OH$ and $CO + OH = CO_2 + H$ channels which are respectively principle to the radical chain branching and heat generation processes of the flames.

The chemical mechanisms that result in hydroperoxy radical production are distinct for each fuel (see Fig. 21).

Methyl butanoate shows a propensity to produce formaldehyde early in the flame structure leading to HO_2 formation through the reaction $HCO + O_2 = CO + HO_2$. The prescription of formaldehyde production is suggested to be the presently limiting process to quantitative accuracy of methyl ester kinetic models against diffusion flame phenomena. It is postulated that the inhibitive effect observed for methyl butanoate on n-alkane diffusion flames will also be observed for more biodiesel like, larger n-alkyl esters.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2011.09.016](https://doi.org/10.1016/j.combustflame.2011.09.016).

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